Replaying both transfer kernels with a second group engine

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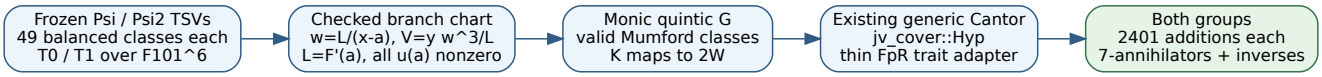
The exported Psi and Psi2 kernels now pass group-law replay with the pre-existing generic Cantor implementation from jv_cover.rs. All49 classes in each corpus are converted to a checked monic quintic chart. The second engine agrees on all2,401 sums per group, all7-annihilators, inverses, validity and distinctness. Complete standalone03 passes8 tests and strict Clippy;PCC9ebcf7. This is correctness evidence without a timing claim. [pcc:unverified]



The missing evidence this adds

The earlier separate flat-field replay checked divisor validity and Frobenius, while additions used the producer’s balanced arithmetic. This new replay uses an existing second polynomial/Cantor implementation with a thin field trait adapter. It imports no producer tower, balanced addition, Richelot map or elliptic kernel generator. The fixed input curves are T0=[31,75,74,54,71,55,1] andT1=[95,26,19,27,94,91,1]. [pcc:unverified]

Source custody and a retained failure

The engine comes from crypto:src/cryptanalysis/jv_cover.rs, last changed at cfa4ef08bccfb509c89f8d0bb0c6c5baf662ff11. The full source SHA256 is 1ecd62c4282402e3411635e0487d0bb36a5d57e15a547f352a7cab1fda7309cc. Retained source ranges40–52 and464–690 contain the generic field trait, polynomials and odd-model Cantor operations. Initial byte-exact source01 and its two-test receipt42b1b7 remain frozen. [pcc:unverified]

StrictClippy02 rejected the old extended-gcd tuple return signature’s type complexity. Final03 uses the identical tuple through a Bezout alias; every arithmetic body stays unchanged. An exact transformation guard permits only that alias declaration and signature substitution. Both source hashes are checked in CI. No lint exemption or reduced test is used; failed02 and full03 are retained,PCC647a5a/9ebcf7. [pcc:unverified]

Why the coordinate bridge preserves divisor classes

A monic quintic over the existing field

For $y^2=F(x)$, choose the first sorted branch a avoiding every displayed support and putL=F’(a). Smoothness makesL nonzero. Set $H(z)=z^6F(a+1/z)$,$w=L/(x-a)$,$V=yw^3/L$ and$G(w)=L^4H(w/L)$. ThenH has degree5 and leadingL,soG is monic degree5 without an ordinate square-root assumption. The inverse is$x=a+L/w$,$y=VL/w^3$. Native checks cover every denominator, degree, leading coefficient and square-freeness. [pcc:unverified]

For$u=x^2+b$ $x+c$ and$A=u(a)$, the transformed support is $U=w^2+L(2a+b)w/A+L^2/A$. The ordinate is v1 $w^2+v(a)w^3/L$ moduloU. Exact polynomial identities prove these formulas in the whole finite support algebra, including multiplicities;no support points are individually chosen. Four KernelChart Lean lemmas pass the kernel/axiom gate and statement bindings,PCC2026a6. Their inverse and chart hypotheses are explicit. [pcc:unverified]

Every nonidentity exported class has degree2 and balanced-plus0, hence isD_finite-K. The two old infinities map to$(w0,V=\pm L^2)$. The functionw has divisorBplus+Bminus-2W, whereW is the quintic’s unique Weierstrass infinity. ThusK maps to2W up to linear equivalence, and the ordinary Mumford representative has exactly the correct baseline. Projective completion and Picard functoriality remain written in INDEPENDENT_GROUP_REPLAY_V2.md/PCC40a568. [pcc:unverified]

Limits and next work

The replay covers the two exact49-class corpora and identity. It does not independently evaluate either isogeny, establish map-kernel membership, replay every arbitrary Jacobian class, or supply a signed scientific verdict. Kernel identification still uses the written global commuting square and the producer’s actual elliptic-map killing controls. Prior limited replay claims stay unchanged. [pcc:unverified]

The full consumer CT24 guard separately passes25 native/80 reference unit and five integration tests,one pre-existing ignored benchmark,PCC5383d3. All partial Lean source modules now contain125 written declarations; geometric dependencies remain explicit. CT25 prepares direct C2 norms; its seven cleared/conditional lemmas pass,c2aaaa,while runtime integration requires complete generic comparisons. No measured computational advantage is established. Admission/livebudget and vault/ledger constraints persist. [pcc:unverified]

Reproduction:successor-preflight/legacy.sha256,src/kernel_group_replay.rs, source extraction guard,fixtures/explicit-psi-kernel-07.tsv and explicit-psi2-kernel-12.tsv. Raw receipts and source snapshots are in `evidence/second-cantor-*`. Existing group-engine source and the written chart/Picard proof state all assumptions and remaining scope.